Lesson 2 Color Tracking

1. Program Logic

The overall implementation process of this lesson is as follow.

First, program ArmPi FPV to recognize colors with Lab color space. Convert the RGB color space to Lab, image binarization, and then perform operations such as expansion and corrosion to obtain an outline containing only the target color. Use circles to frame the color outline to realize object color recognition.

Secondly, after color recognition, the height of the robotic arm is processed. The X, Y and Z coordinates of the center of the image are taken as the set values, and please input the X, Y and Z coordinates obtained currently to update the pid.

Next, robotic arm begins calculating based on image position feedback. Finally, the coordinate value changes linearly with the change of position, so as to achieve the color tracking.

The source code of the program is located in: /home/ubuntu/armpi_fpv/src/object_tracking/scripts

```
    return area_max_contour, contour_area_max  # back to the greatest contour area

# initial position
def initMove(delay=True):
    with lock:
        target = ik.setPitchRanges((0, y_dis, Z_DIS), -90, -92, -88)
        if target:
            servo_data = target[1]
            bus_servo_control.set_servos(joints_pub, 1500, ((1, 200), (2, 500), (3, servo_data['servo3']), (4, servo_data['servo4']), (5, servo_data['servo5']),(6, servo_data['servo6'])))
    if delay:
        rospy.sleep(2)

def turn_off_rgb():
    led = Led()
    led.index = 0
    led.rgb.r = 0
    led.rgb.g = 0
    led.rgb.b = 0
    rgb_pub.publish(led)
    led.index = 1
    rgb_pub.publish(led)
```

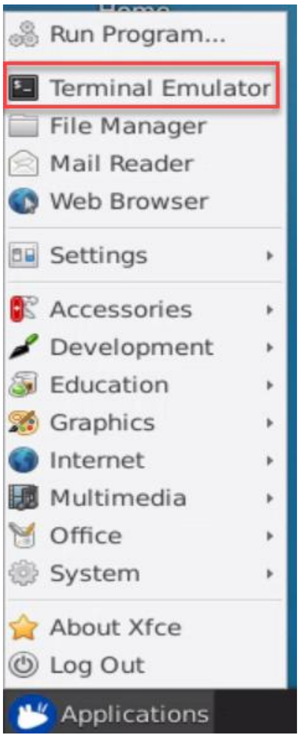
2. Operation Steps

The entered command must be strictly distinguished between uppercase and lowercase and spaces, and the keywords support the "TAB" key completing.

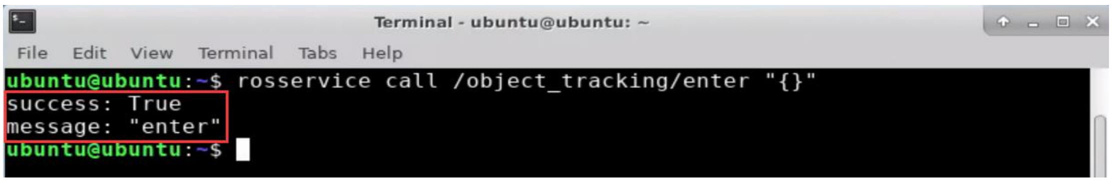
2.1 Enter the Game

① Turn on ArmPi FPV and connect to Raspberry Pi desktop through NoMachine.

② Click “Applications” button and select “ Terminal Emulator” in pop-up interface to open the command line terminal.



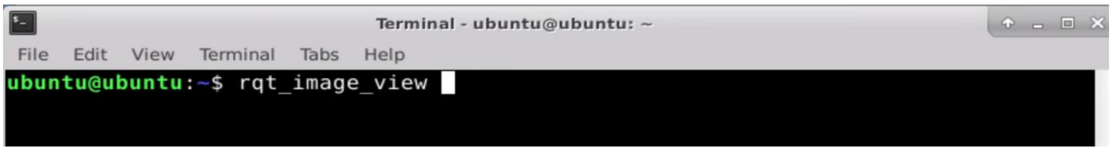
③ Enter “rosservice call /object_tracking/enter "{}"" command, then press “Enter” to start the game. After the game starts, a print prompt will appear, as shown in the figure below.



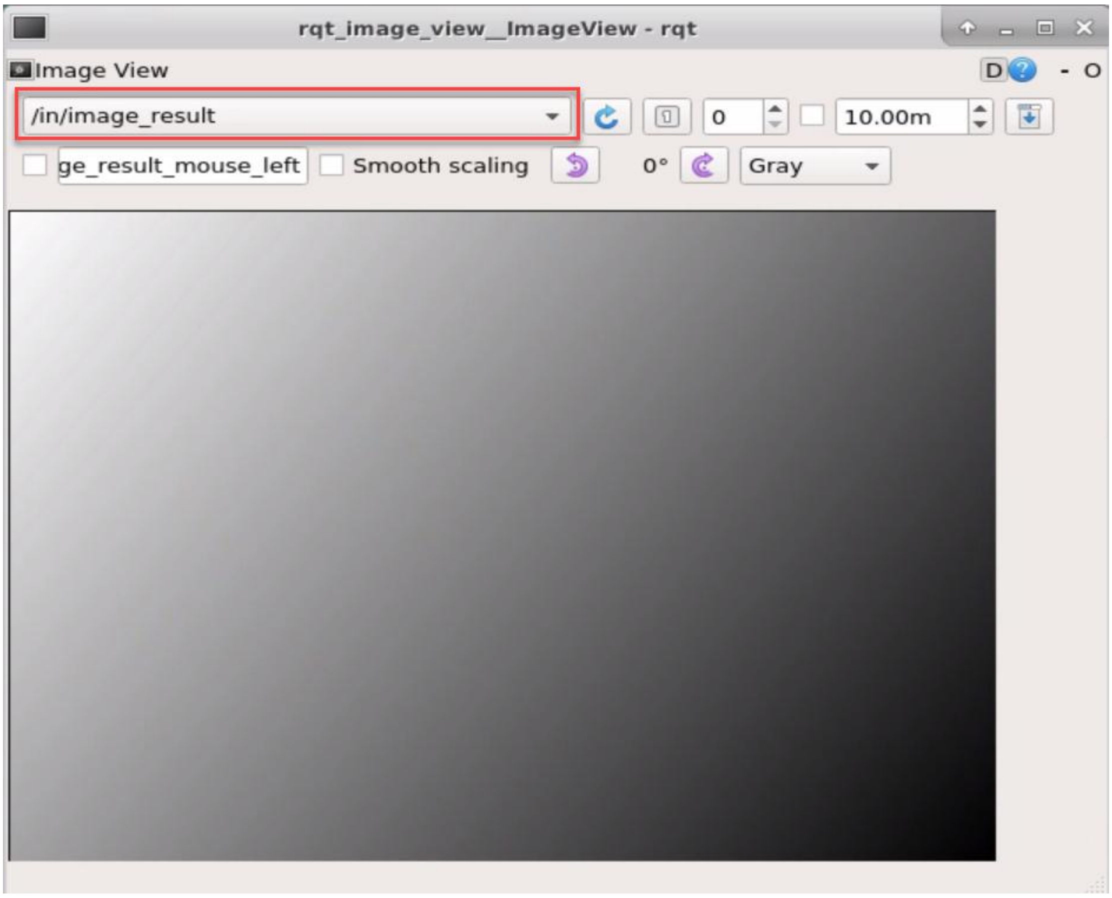
2.2 Turn On Image

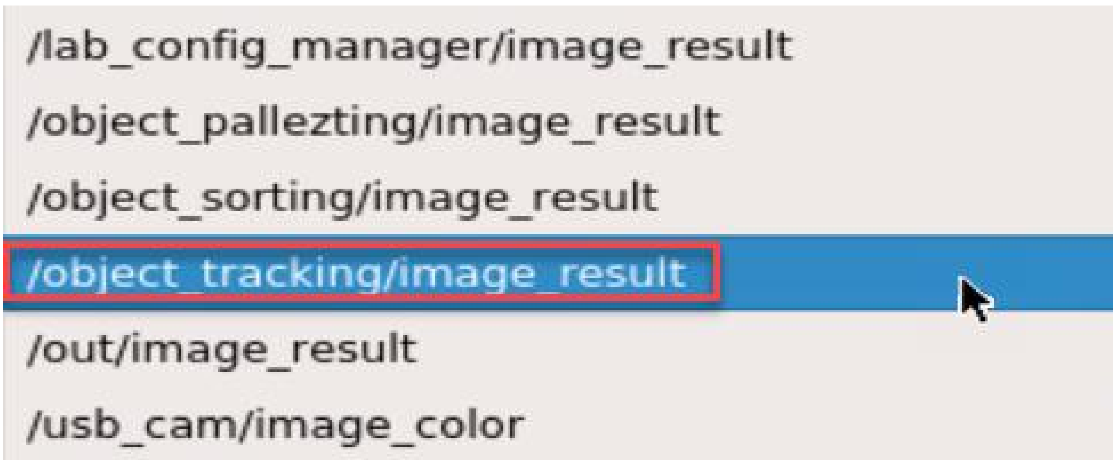
① When entering the game, we need to turn on image returned by camera with rqt. Please go back to the desktop and click the “Application” icon at bottom-left corner to open a new terminal.

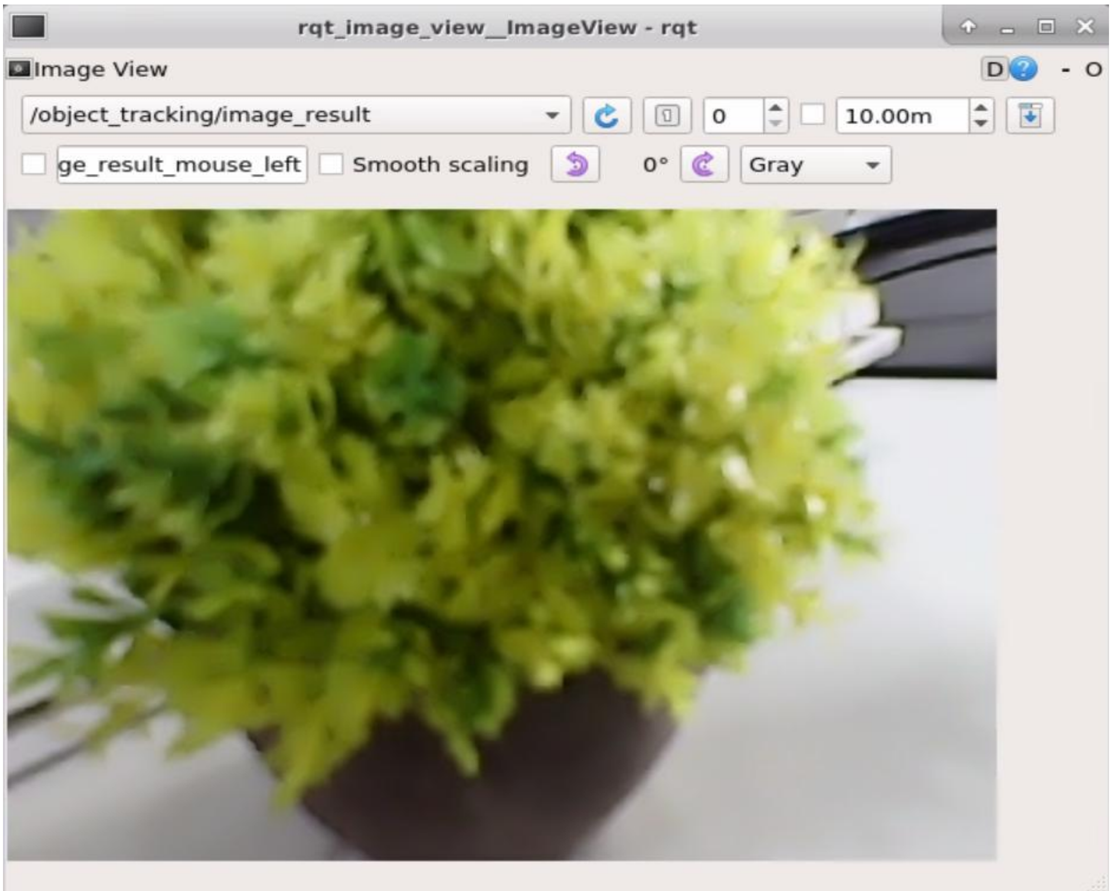
② Enter “rqt_image_view” command and press “Enter”. Then open rqt after few seconds.



③ As shown in the following figure, select "/object_tracking/image_result" and other settings remain the same.







Note: after the image is turned on, please be sure to select the topic corresponding to the game you play. Otherwise, the recognition process can not be displayed normally if you start other games.

2.3 Start the game

Please go back to the terminal opened in “2.1 Enter the Game” and enter “rosservice call /object_tracking/set_running "data: true"” command. When the prompt, shown in the red box, appears, it means you start the game successfully.

```
ubuntu@ubuntu:~$ rosservice call /object_tracking/set_running "data: true"
success: True
message: "set running"
ubuntu@ubuntu:~$
```

After starting the game, we need to select the targeted color. If the targeted color is red, please enter “rosservice call /object_tracking/set_target "data: 'red'"” command.

If the targeted color is green or blue, please change the word “red” into green or blue. (Please strictly distinguish between uppercase and lowercase.)

```
ubuntu@ubuntu:~$ rosservice call /object_tracking/set_target "data: 'red'"
success: True
message: "set_target"
```

2.4 Pause and Quit the game

You can enter “rosservice call /object_tracking/set_running "data: false"” command to pause the game. Referring to “2.3 Start the game”, you can change the targeted color into green or blue after a pause.

```
ubuntu@ubuntu:~$ rosservice call /object_tracking/set_running "data: false"
success: True
message: "set_running"
ubuntu@ubuntu:~$
```

You can enter “rosservice call /object_tracking/exit "{}"” command to quit the game.

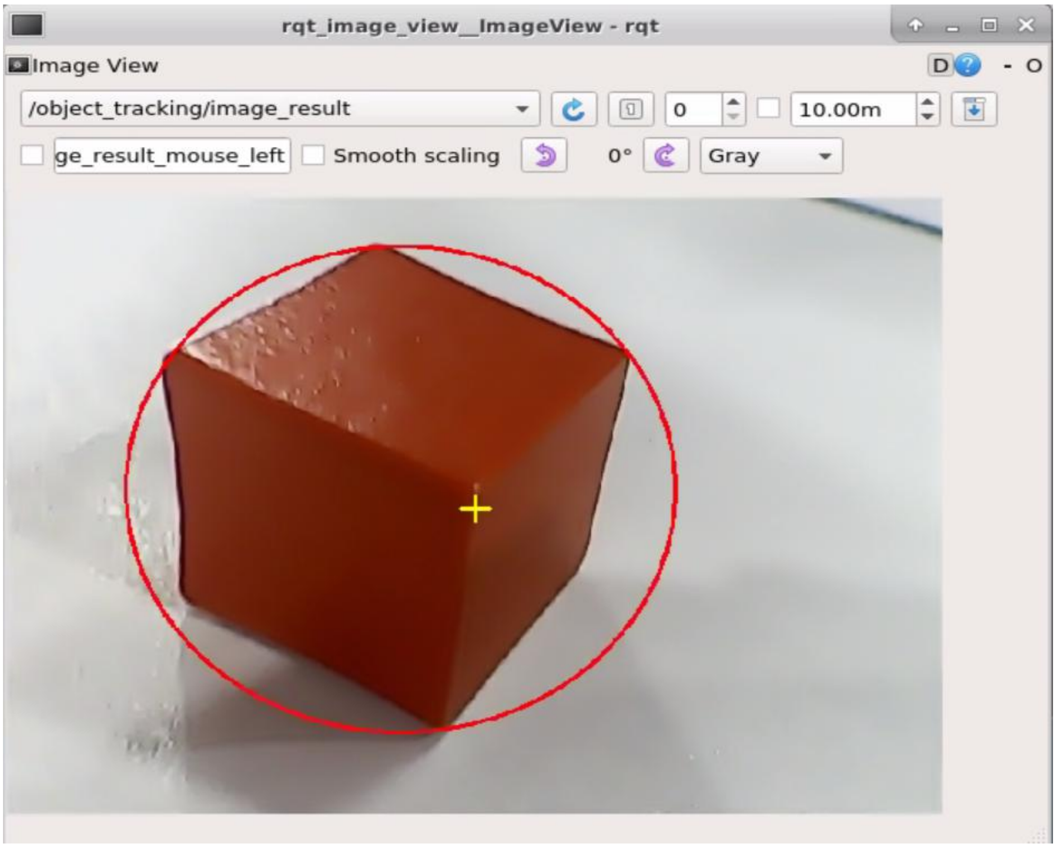
```
ubuntu@ubuntu:~$ rosservice call /object_tracking/exit "{}"
success: True
message: "exit"
ubuntu@ubuntu:~$
```

Note: the current game will continue as Raspberry Pi is powered on. To avoid occupying too much RAM, please quit the current game according to the instruction above before playing other AI vision game.

If you want to turn off the image returned by camera, you can turn back to open rqt terminal and press “Ctrl+C”.

3. Function Realization

Place the red block in the vision recognition area. When the red block is recognized, the screen returned by the camera will circle the red block. Hold the red block to move slowly and robotic arm will move to chase it.

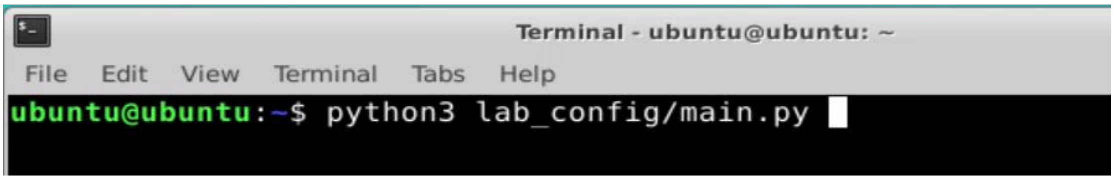


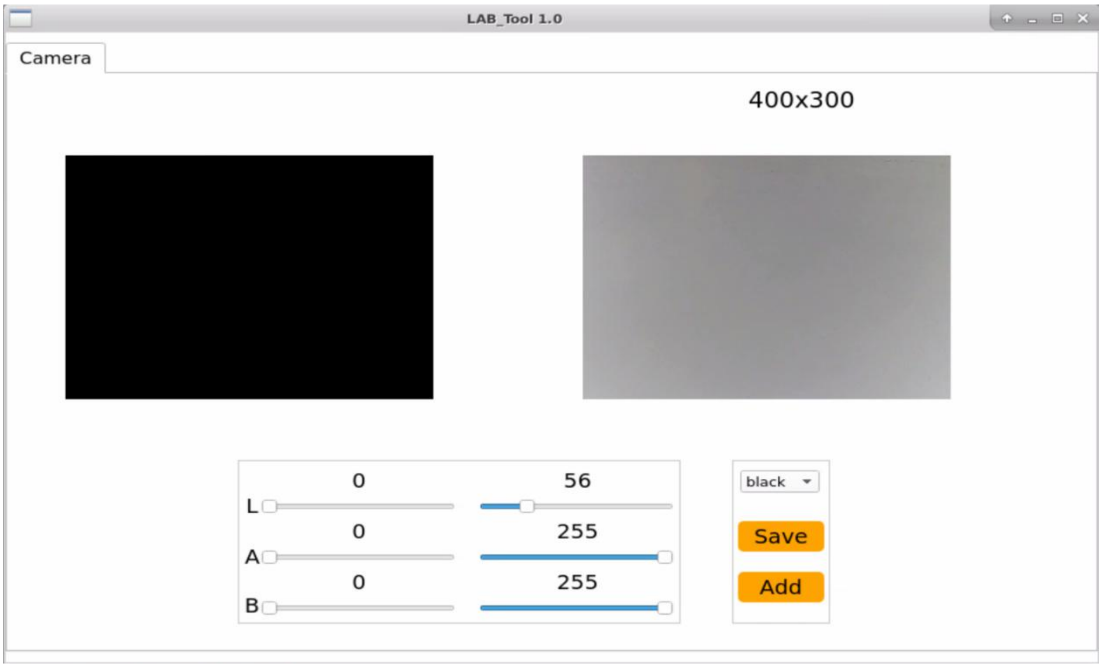
4. Function Extension

4.1 Add New Recognition Colors

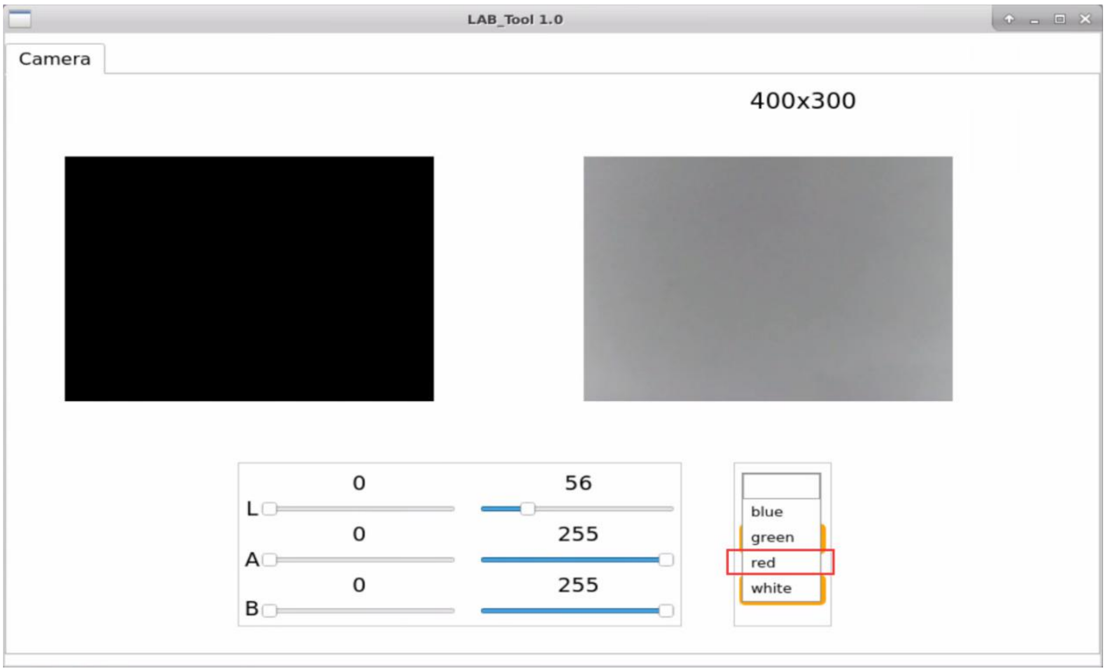
There are three built-in recognition colors in the color tracking program, including red, green and blue. Besides these three colors, you can add new recognition colors. Let's take orange for example.

① Please open the terminal interface and enter "python3 lab_config/main.py" command to open color threshold debugging tool. If there is no image returned by camera in the pop-up interface, it means the camera has not been connected successfully. And you should go back to check the camera connection.



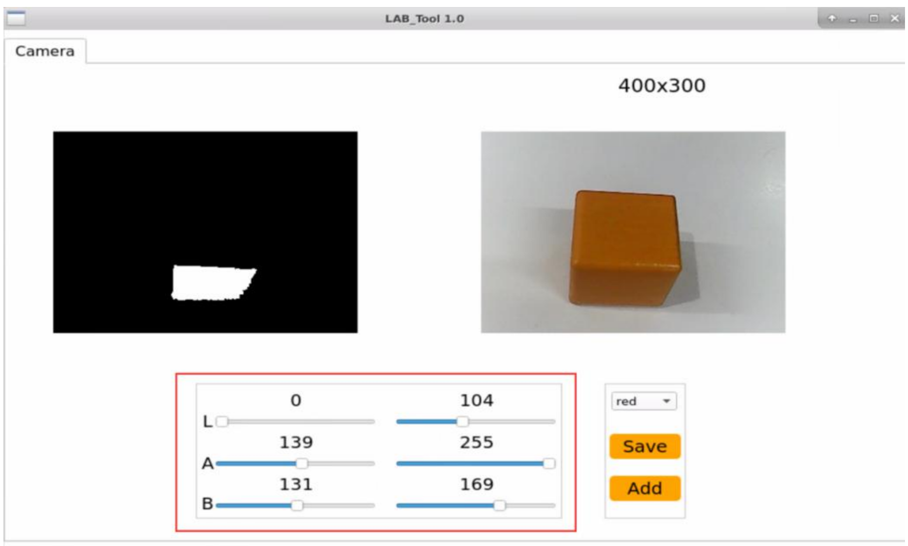


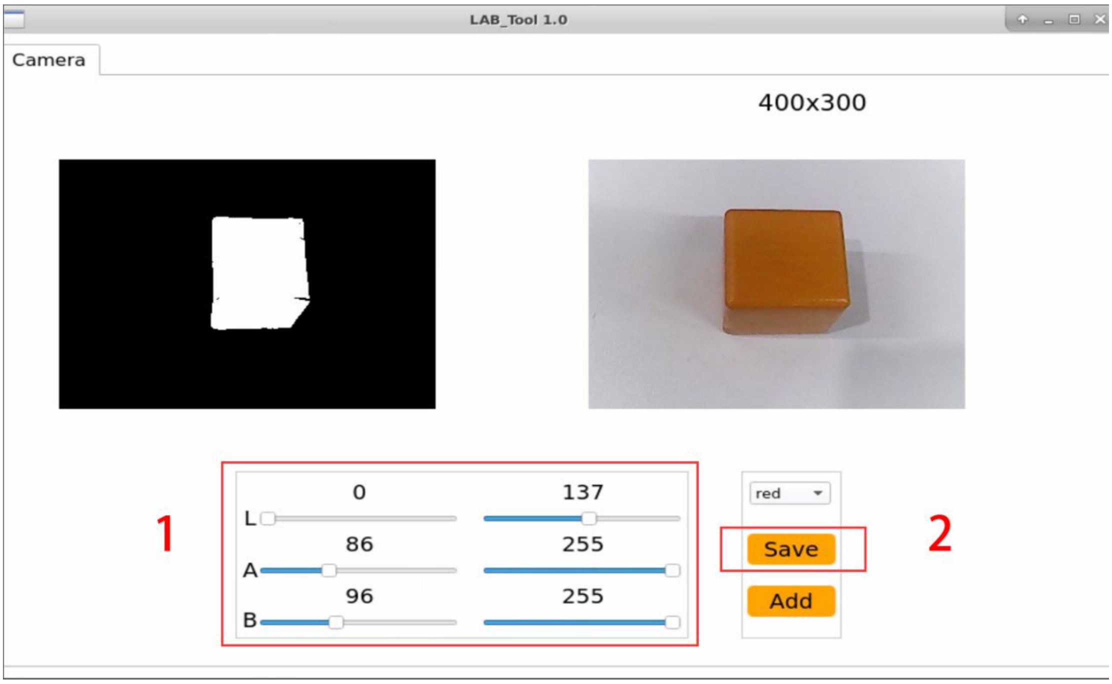
② After connecting successfully, please select "red". (As shown in the picture)



③ At this time, the image returned by camera is on the right of the interface and the box to recognize color is on the left. Focus the camera on yellow item, then drag the six sliders below until the orange part of the left screen becomes white and area outside of orange become black, and then click " Save" button to keep the modified data.

It is recommended to use screenshot to record the initial value, which is beneficial for you to change the value back. Or you can follow step ③ to set red as the recognition color.





④ Please follow the first three steps in “2. Operation Steps”, from 2.1 to 2.3, to play the color tracking games.

In the previous steps, we have saved orange as red, therefore we should select the targeted color to “red” before starting the game.

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⑤ Place the orange item in front of the camera and hold it to move slowly. The ArmPi FPV will follow the movement of the item. If you want to modify to other colors, please operate as the above steps.